CHARGING TECHNIQUES OF EV BATTERY



PROBLEM STATEMENT

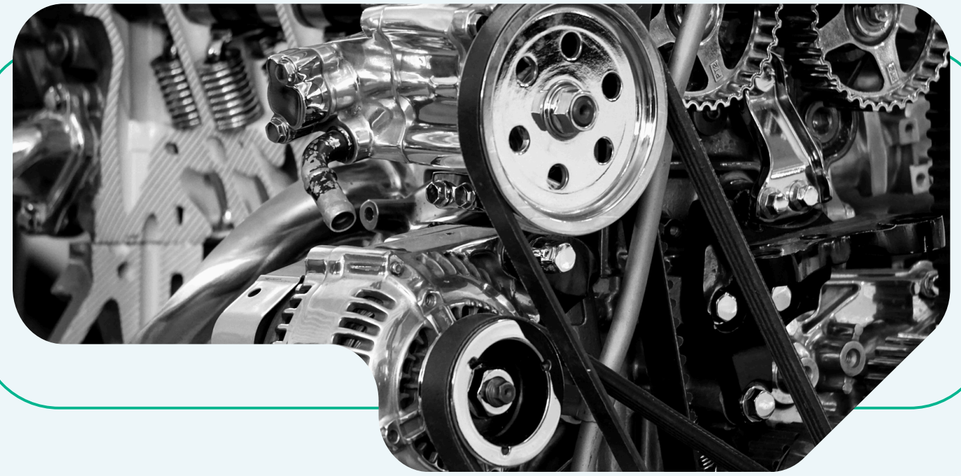
To design and simulate a battery charging system using different charging methods to achieve fast ,efficient, and safe , charging of lithium ion battery



INTRODUCTION

- Battery charging significantly affects the performance, efficiency, and lifespan of rechargeable batteries used in applications like electric vehicles and portable devices.
- Various charging techniques such as CC, CV, CC–CV, and temperature-based methods are used to balance charging speed and safety.
- Each method has distinct characteristics in terms of voltage and current behavior, along with specific advantages and limitations.

MOTIVATION



- **The increasing use of rechargeable batteries in EVs and electronics requires efficient and safe charging methods.**

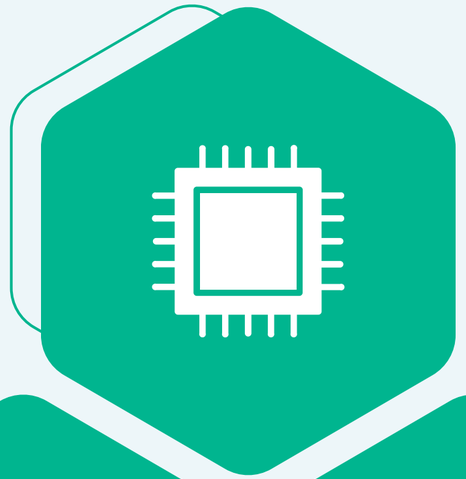
Different charging techniques affect temperature, speed, safety, and battery life in different ways.

Understanding these effects is important to improve battery performance and reliability.

OBJECTIVES

Temperature

To analyze how different charging techniques affect battery temperature and thermal behavior during charging.



Safety

To evaluate the safety of each charging method in terms of overcharging, overheating, and risk of battery damage



Speed

To compare the charging speed of different techniques by measuring the time taken to reach full charge



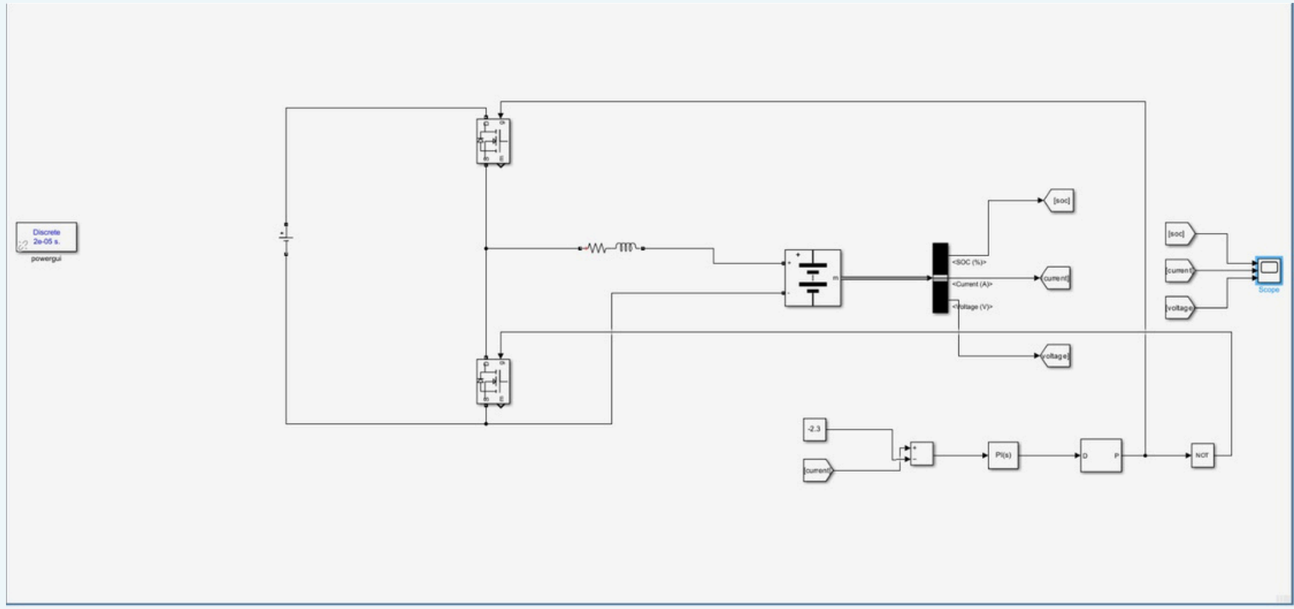
Battery Life

To study the impact of charging methods on battery health and overall cycle life.

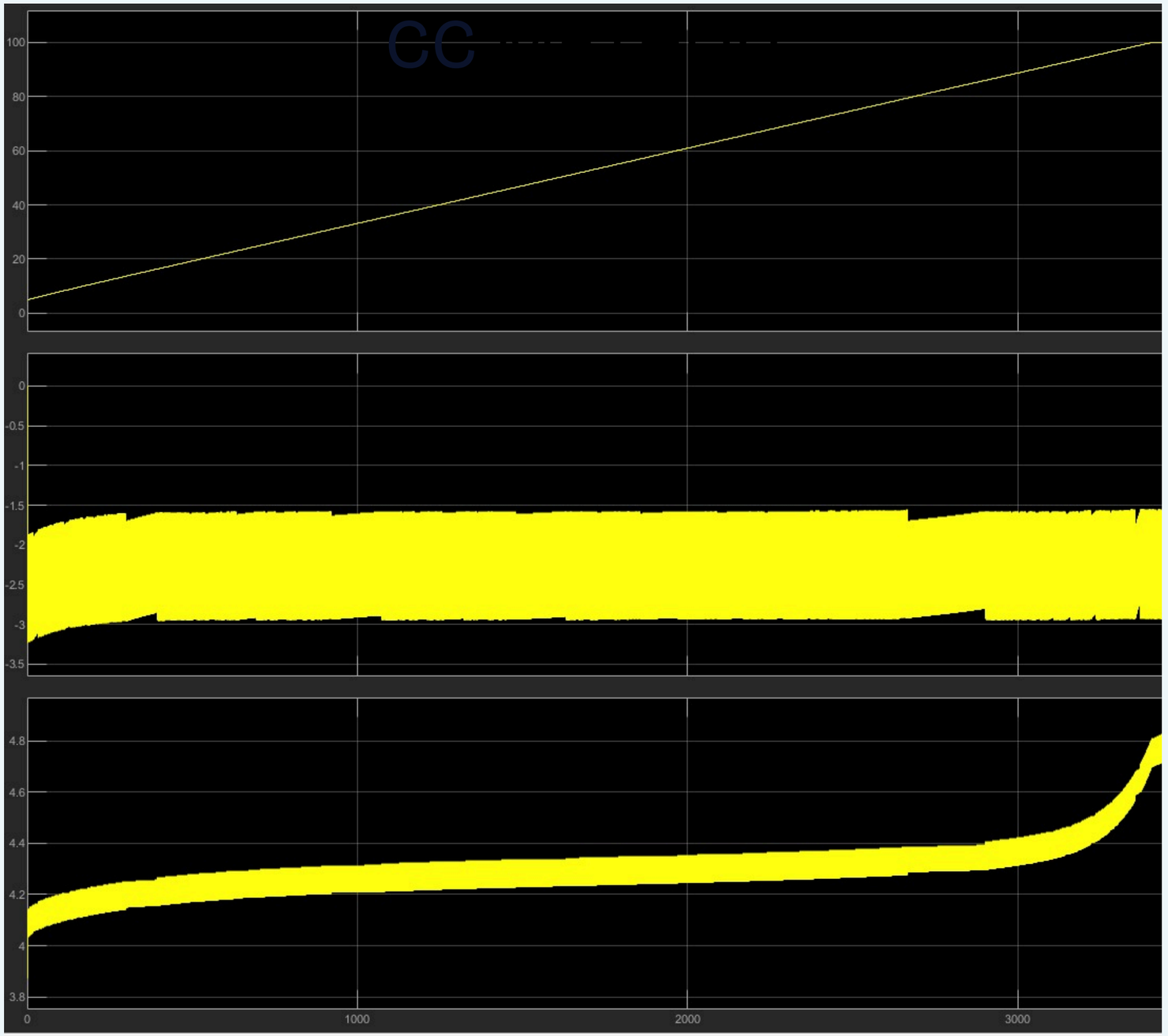


CC METHOD

Constant Current (CC) charging means supplying a steady current to the battery so it charges quickly, especially when it’s low on charge. As the battery fills up, its voltage rises.



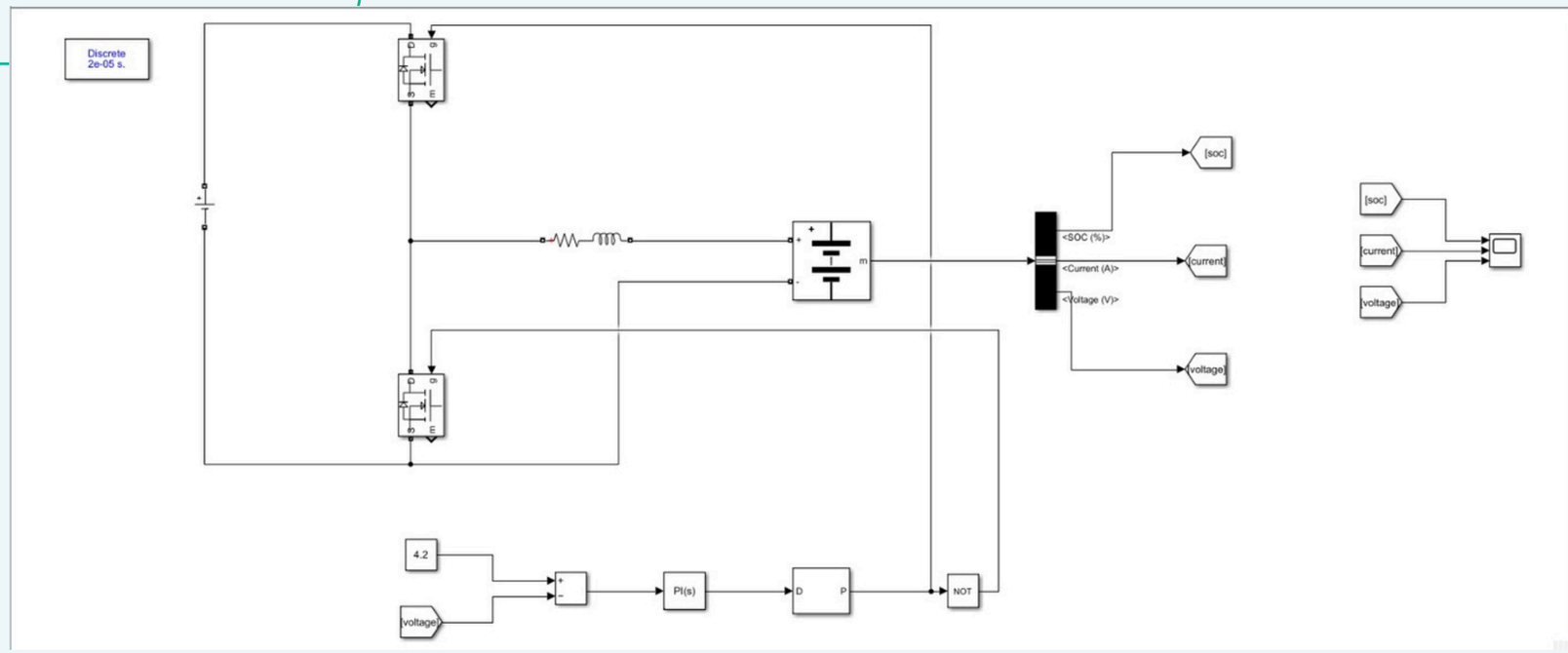
- We observed that In constant current charging, the battery charges very rapidly without proper voltage control.
- This is causing the battery voltage to exceed its safe limit during charging.
- It leads to overcharging, overheating, and reduceing battery life.


cc
100
80
60
40
20
0
0
-0.5
-1
-1.5
-2
-2.5
-3
-3.5
4.8
4.6
4.4
4.2
4
3.8
0
1000
2000
3000

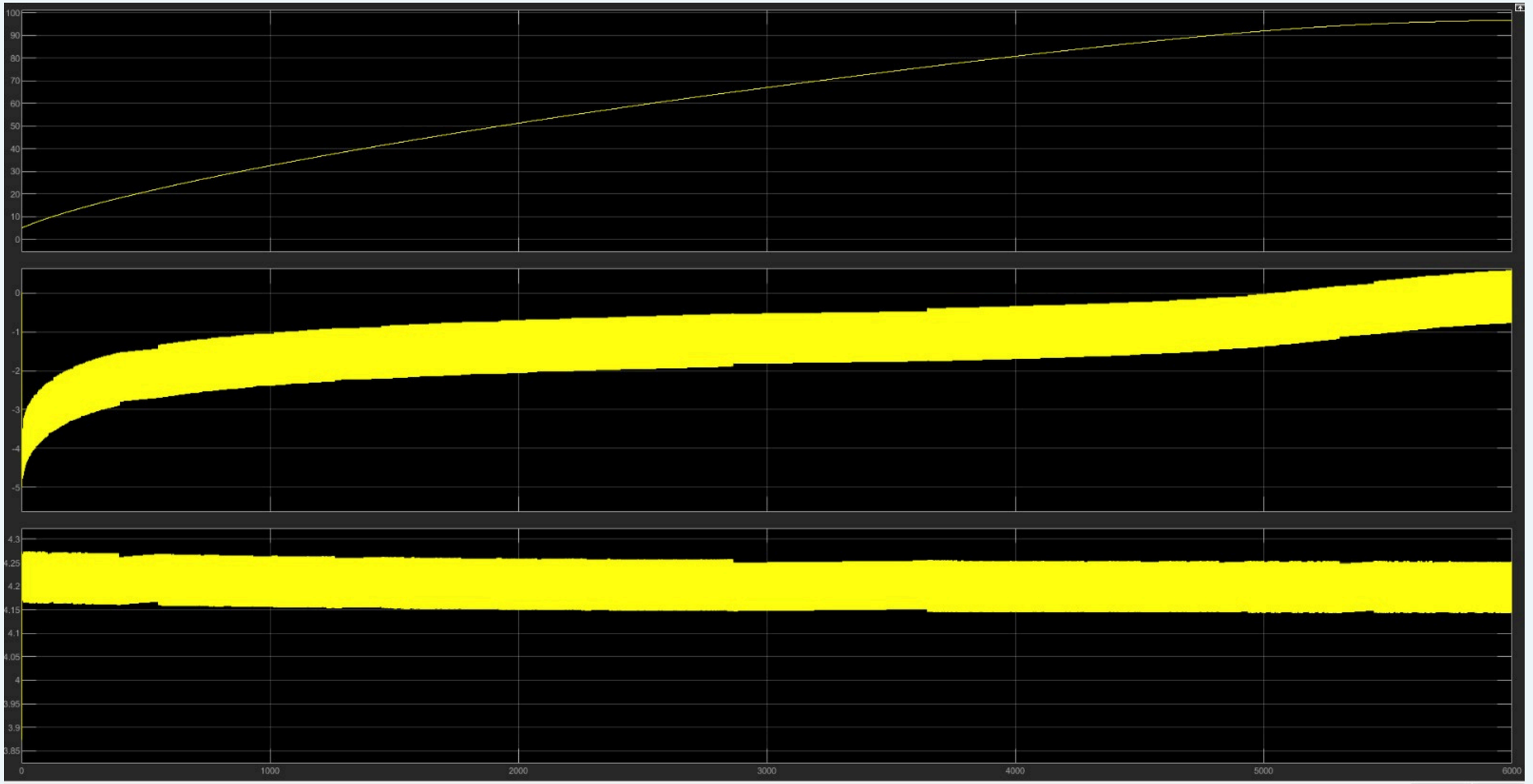
CV METHOD

Constant Voltage (CV) charging maintains a fixed voltage across the battery during charging. As the battery charges, the current gradually decreases while the voltage remains constant.

- We observe that, In constant voltage charging, the voltage is maintained fixed, causing the charging current to gradually decrease.
- This results in a very slow charging process, so it takes long time to reach 100% state of charge (SOC).



CV METHOD


100
90
80
70
60
50
40
30
20
10
0
0
-1
-2
-3
-4
-5
4.3
4.25
4.2
4.15
4.1
4.05
4
3.95
3.9
3.85
0
1000
2000
3000
4000
5000
6000

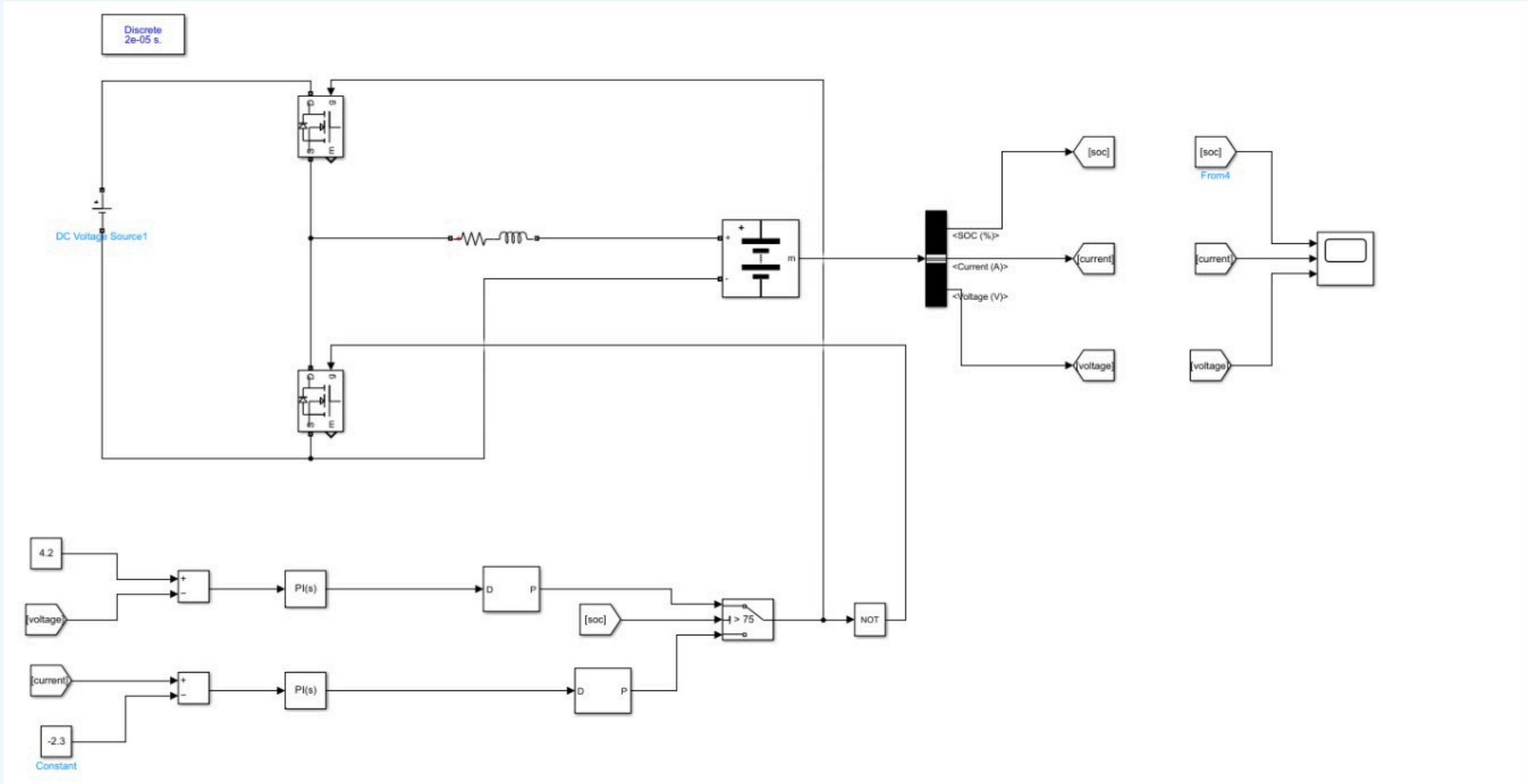
CC-CV METHOD

CC–CV charging is a two-stage method that combines constant current and constant voltage charging. In the first stage, the battery is charged with a constant current, causing the voltage to gradually increase. Once the battery reaches its set maximum voltage, the method switches to constant voltage mode. In this second stage, the voltage is kept constant while the current slowly decreases. This approach ensures efficient charging in the beginning and safer, controlled charging as the battery reaches full capacity.

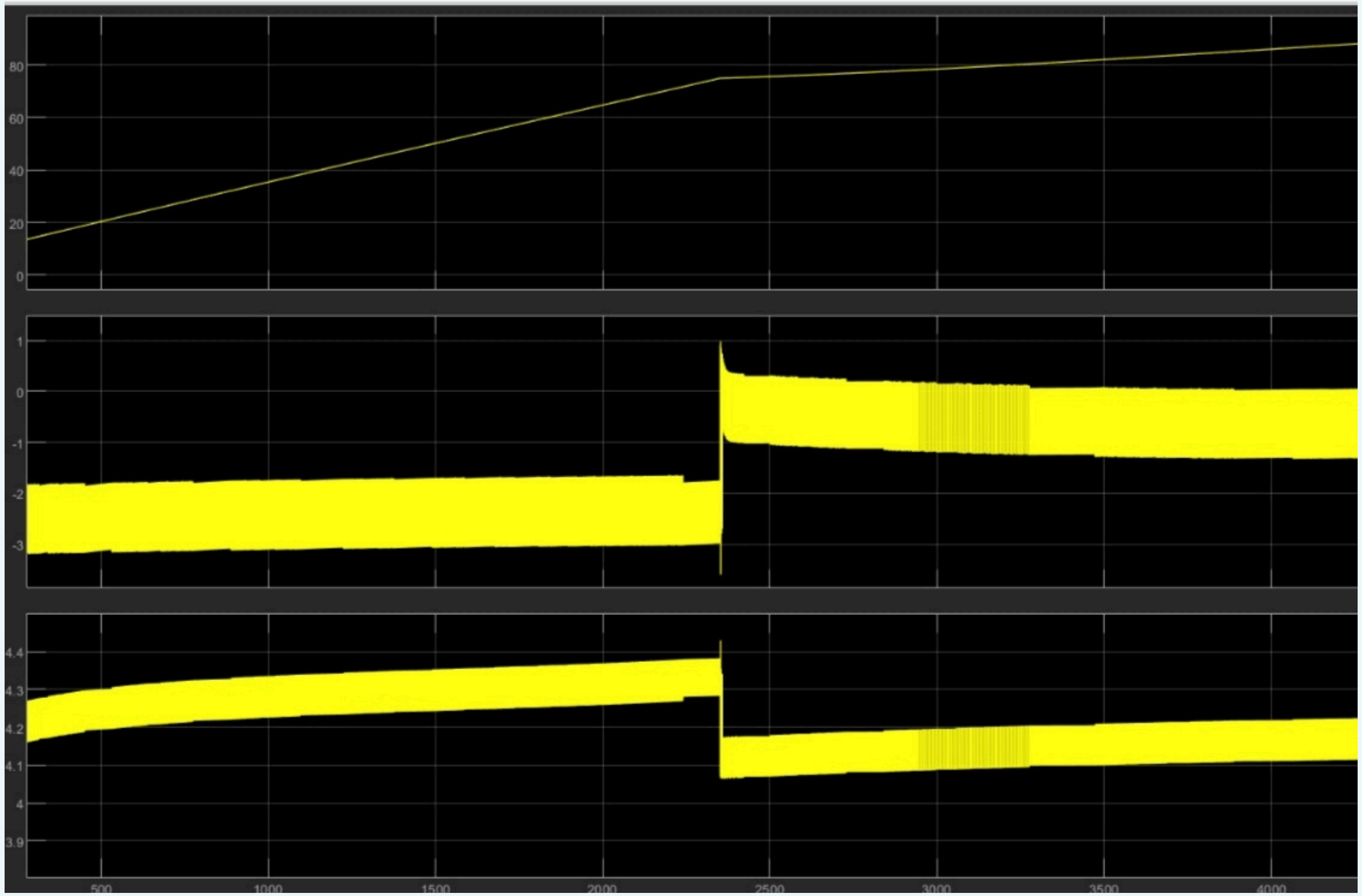
In this method we are able to control the overcharging and can reduce the charging time

But the charging control is more complex, requiring precise switching between constant current and constant voltage modes.

CC-CV METHOD


Discrete
2e-05 s.
DC Voltage Source1
[soc]
[current]
[voltage]
<SOC (%)>
<Current (A)>
<Voltage (V)>
From4
4.2
PI(s)
D
P
> 75
NOT
-2.3
Constant

CC-CV METHOD

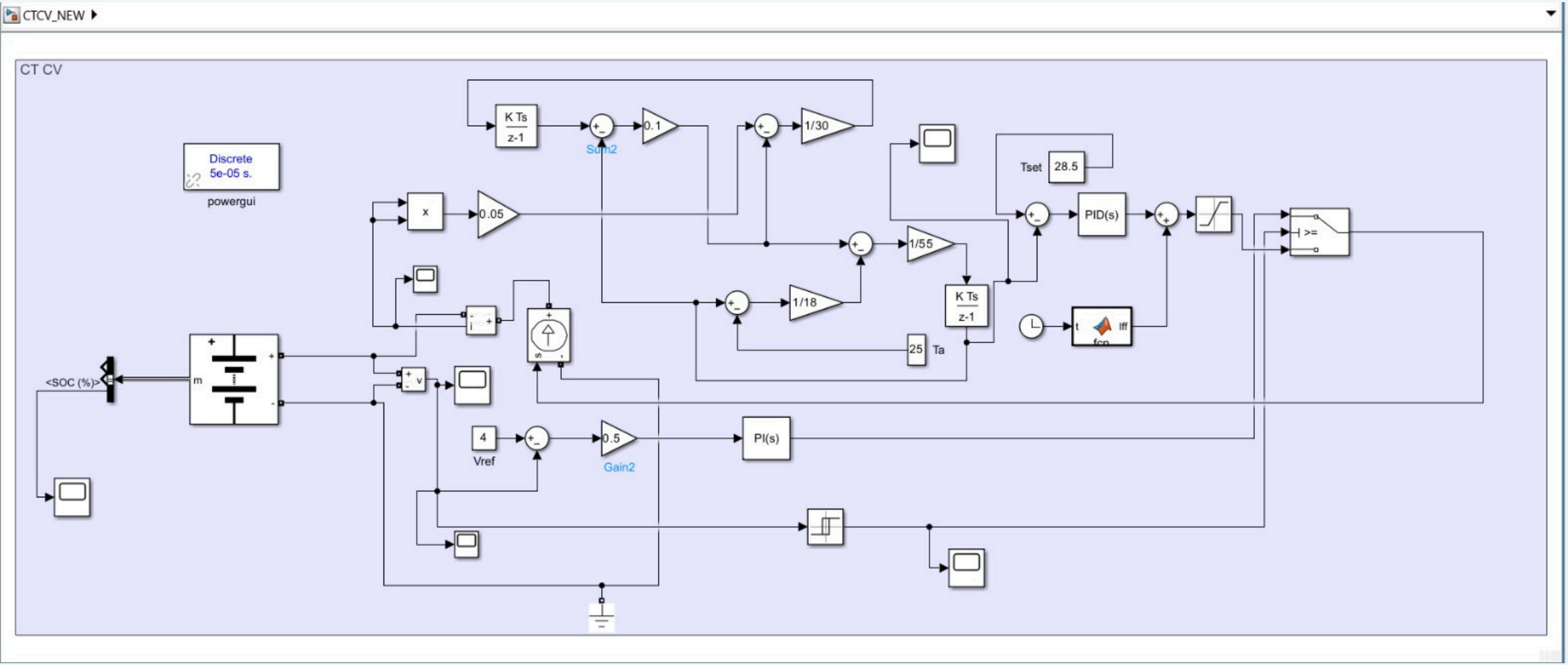

80
60
40
20
0
1
0
-1
-2
-3
4.4
4.3
4.2
4.1
4
3.9
500
1000
1500
2000
2500
3000
3500
4000

CT-CV METHOD

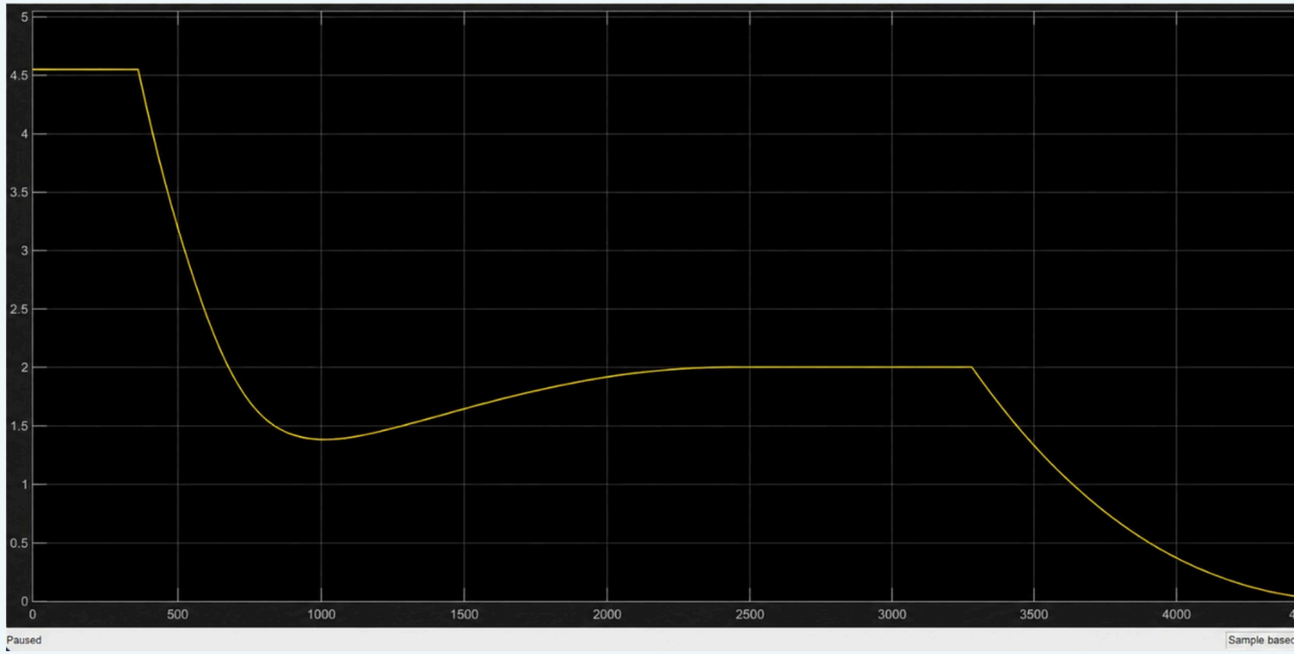
CT–CV charging combines temperature-based control with constant voltage charging. In the initial stage, the charging current is adjusted based on the battery’s temperature to prevent overheating. As the battery approaches its safe voltage limit, the method switches to constant voltage mode. In this stage, the voltage is held constant while the current gradually decreases

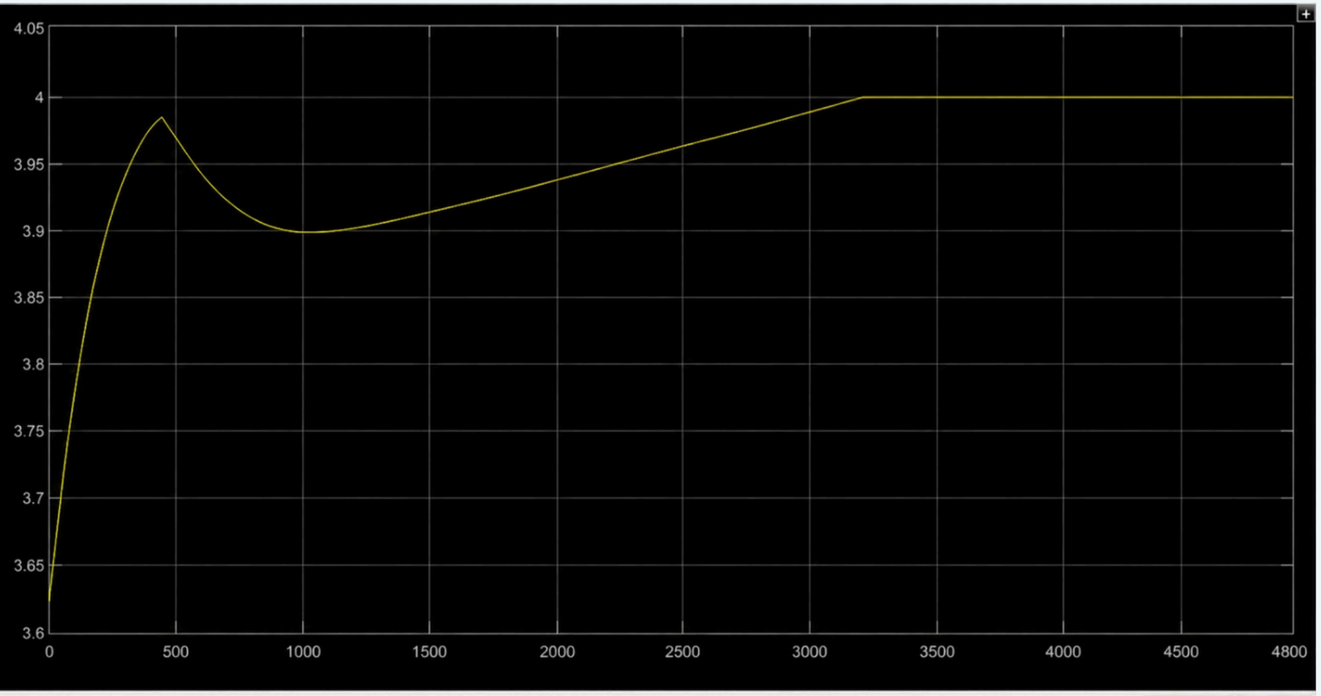
- In CT–CV charging, a thermal model is developed to monitor battery behavior during charging. This model helps in observing both internal temperature and surface temperature in real time.

CT-CV METHOD


CTCV_NEW
CT CV
Discrete
5e-05 s.
powergui
K Ts
z-1
Sum2
0.1
1/30
x
0.05
1/55
1/18
Tset
28.5
PID(s)
25
Ta
<SOC (%)>
m
4
Vref
0.5
Gain2
PI(s)
t
fcn
Iff
>=

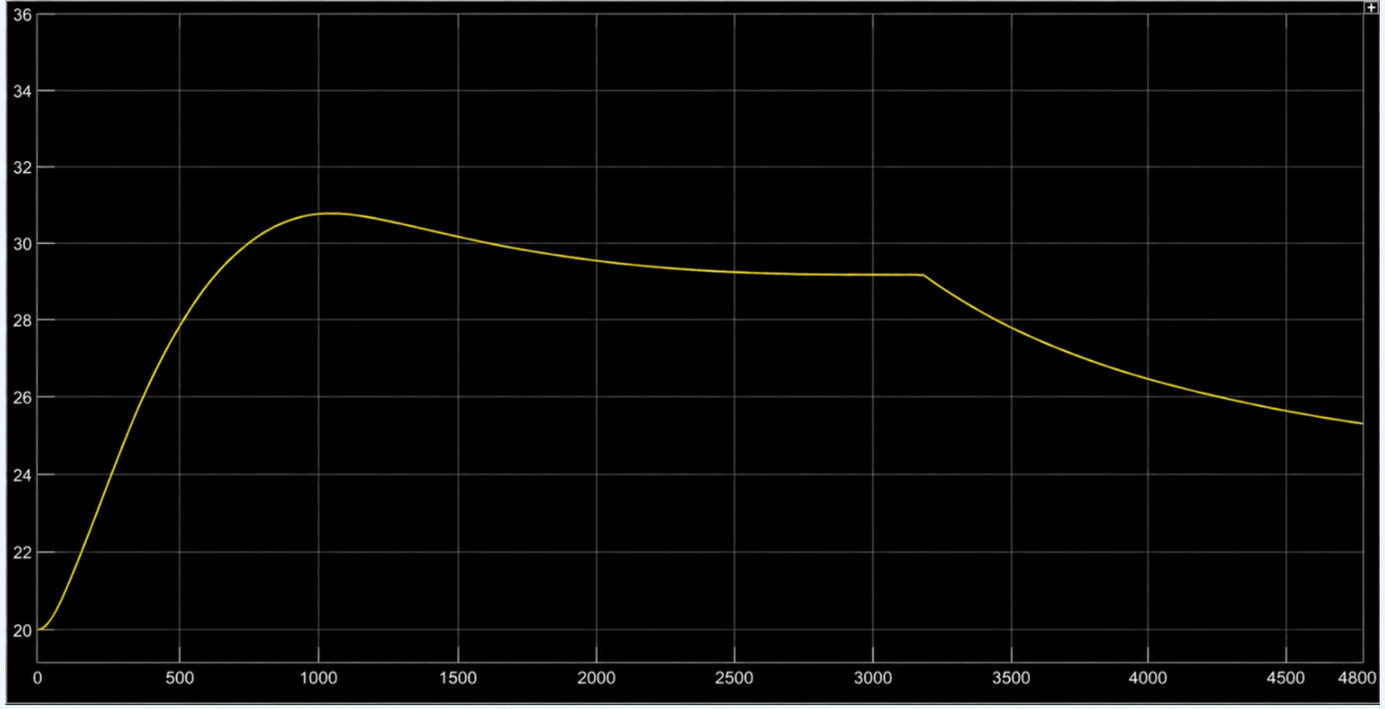
CURRENT Vs TIME

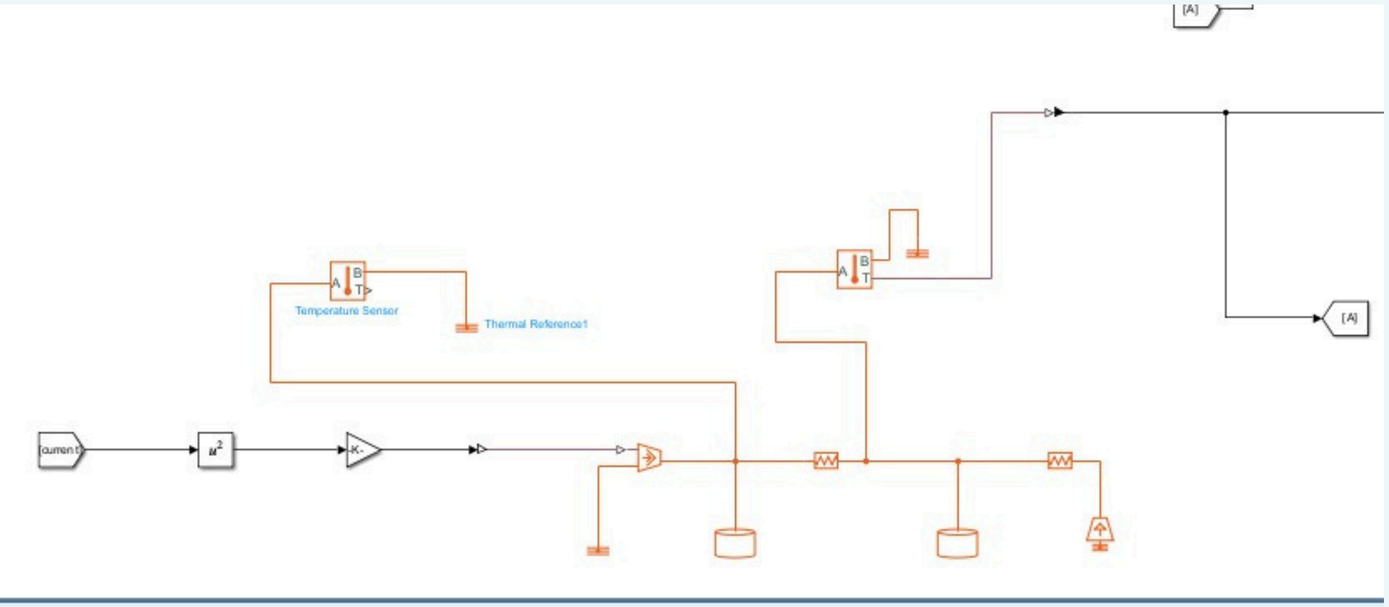




Vbattery Vs TIME

SURFACE TEMP Vs TIME

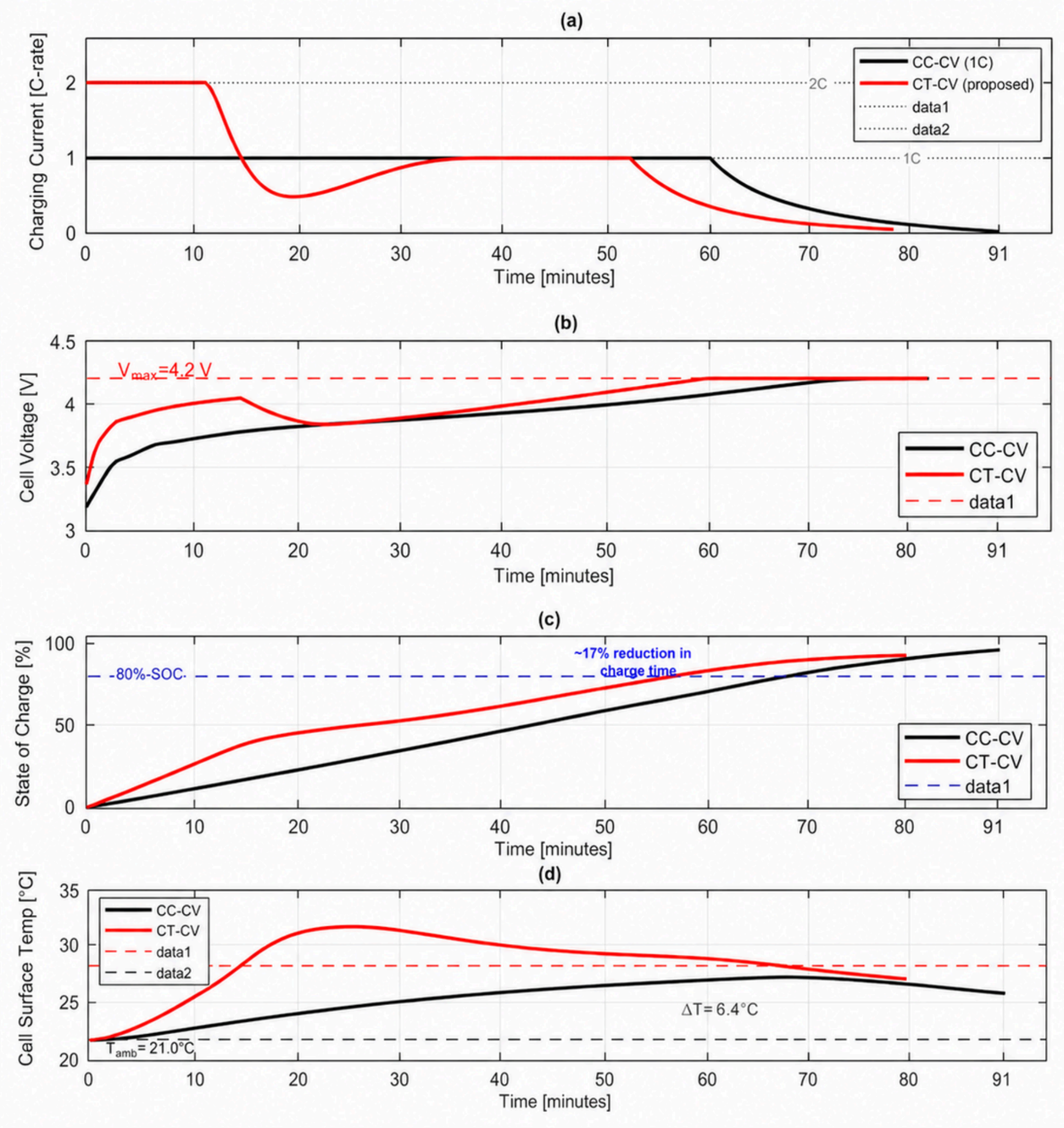




THERMAL MODEL USING REAL COMPONENTS

RESULTS

METHOD	CHARGING TIME	TEMPERATURE RISE
• CC	61 min	
• CV	104 min	
• CC-CV	91 min	• ΔT=6.5°C
• CT-CV	80 min	• ΔT=6.5°C



CONCLUSION

- The proposed charging profile reduces the overall charging time compared to the conventional CC–CV method.
- It effectively limits the temperature rise by incorporating thermal control during charging.
- Thus, the CT-CV method ensures faster, safer, and more efficient battery charging performance.

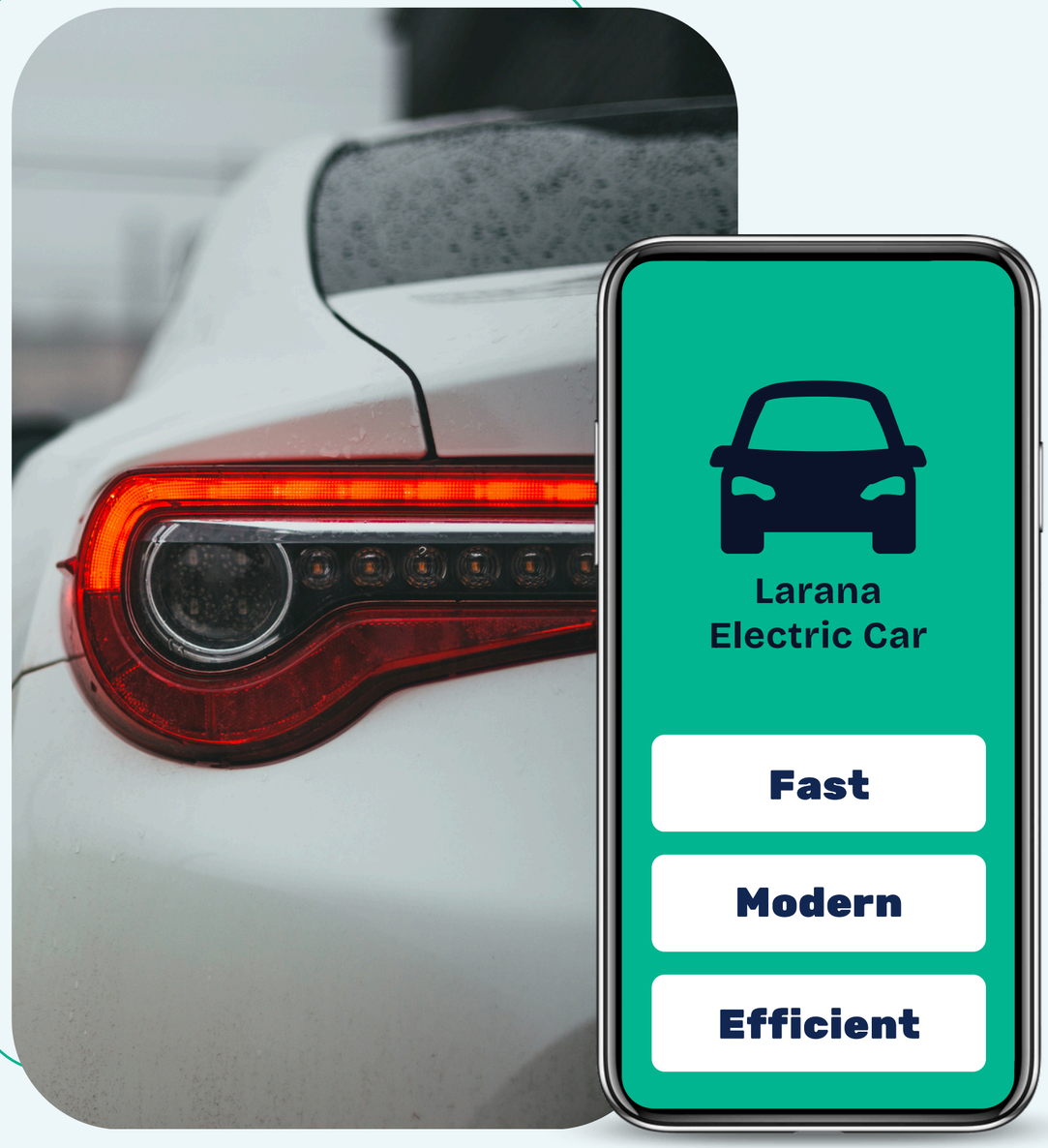


REFERENCE:

IEEE PAPERS:

A Closed-Loop Constant-Temperature Constant-Voltage Charging Technique to Reduce Charge Time of Lithium-Ion Batteries

Performance Analysis of the Lithium-Ion Batteries using CT-CV Charging Technique



Thank You!

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